MD. ABU HORAIRA

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Portfolio | LinkedIn

EDUCATION

University of Dhaka Dhaka, Bangladesh
Bachelor of Science in Computer Science & Engineering August 2021 – January 2027 (Expected)
CGPA: 3.80/4.00 Till 3rd Year, 2nd Semester

Comilla Victoria Government College
Higher Secondary Certificate (Science), 2021
GPA: 5.00/5.00

Comilla Shikkha Board Model College
School Secondary Certificate (Science), 2019
GPA: 5.00/5.00

TECHNICAL SKILLS

Programming: C, C++, Java, Python, JavaScript, TypeScript, Dart
Web & Backend: React.js, Node.js, Express.js, HTML, CSS
Databases: MongoDB, MySQL, Oracle SQL, Firebase
Cloud & Tools: Azure, Docker, Nginx, Git, GitHub, Linux, Android Studio
AI & Embedded: Machine Learning, Jetson Nano, Microcontrollers, UAV/Drone Systems

THESIS(Ongoing)

An Edge-Deployable Agentic AI Framework for Multi-Domain Autonomous UAV Mission Execution with Task-Aware Dynamic LoRA Adaptation
Developing an edge-deployable agentic AI system for autonomous UAV mission execution using natural-language commands, dynamic LoRA capability selection, vision-based context understanding, and mission planning. The system integrates an on-device language model with PX4-based flight control for resource-constrained UAV deployment across agriculture, emergency response, and infrastructure inspection.
Supervisor: Dr. Mosaddek Hossain Kamal Tushar, Professor, CSEDU [Profile]

PROJECTS

CSEDU Nexus – Student Organization Management System GitHub
Built and deployed a full-stack organization management platform using React, TypeScript, Node.js, Express, and MongoDB, implementing JWT/RBAC, event and election management, finance, QR attendance, payments, notifications, and video meetings with Docker, Nginx, and Azure VM.

TravelHeaven – AI-Powered Travel Planning Platform GitHub
Developed a MERN-stack travel platform integrating Google Gemini AI and Bangladesh Railway API for intelligent travel planning and railway information services.

Rescue Track – Disaster Relief Management System GitHub
Built a disaster management platform with five user roles, relief-camp management, inventory tracking, automated replenishment, SSLCommerz payments, and Nodemailer-based communication.

LearnSphere – Cross-Platform EdTech Application GitHub
Developed a cross-platform mobile learning application with courses, video lectures, assignments, quizzes, Firebase Authentication, and Cloud Firestore.

Hospital Management System – Healthcare Management Platform GitHub
Designed a healthcare management system for doctors, patients, and administrators using Node.js and Oracle SQL to support core hospital operations.

CSEDU Bank Management – Desktop Banking Application GitHub
Built a JavaFX desktop banking application with MySQL, authentication, account management, and transaction processing for deposits, withdrawals, and transfers.

DATASET PUBLICATION

An Image Dataset for Illegal Road Crossing Detection in Bangladesh Mendeley Data
Version 1, 2026
Co-authored and curated 694 real-world images from Dhaka, Bangladesh, with YOLO-format annotations for three-class and single-class road-crossing detection.

ADDITIONAL INFORMATION

Interests: Software Engineering, Artificial Intelligence, Machine Learning, Embedded Systems
Languages: Bengali (Native), English (Fluent)